Biomedical Signal Processing and Machine Learning – Module 1 – Practical exam

Second Cycle Degree/Two Year Master In Biomedical Engineering

21 July 2025

Setting up the development platform (Google Colab)

1. Go to eol.unibo.it, log in to today's session, and download the file **'Template_Module1_20250721.ipynb'.**
2. Go to https://colab.research.google.com/ , and log in at the top right with your Google account.
3. Upload the template to Google Colab, for example, by selecting **'File -> Upload Notebook -> Upload'**

Documentation

Directly use the following links for the Python documentation (DO NOT use Google or other search engines because of firewall settings):

- https://pandas.pydata.org/docs/
- https://scikit-learn.org/stable/
- https://numpy.org/doc/
- https://matplotlib.org/stable/index.html

Exercise

You are provided with a dataset generated from an unknown function with added noise. The data is already loaded in the variables X and y in the provided Python template file.

1. Split the dataset into training and test sets.
2. For each polynomial degree from 1 to 10:
   - Train the polynomial model on the training data.
   - Compute the Mean Squared Error (MSE) on both the training and test sets.
3. Create a plot showing training MSE and test MSE as functions of the polynomial degree.
4. Discuss the results

Submitting the solution on eol.unibo.it

Save your Python code on your PC, rename it, and upload it to eol.unibo.it